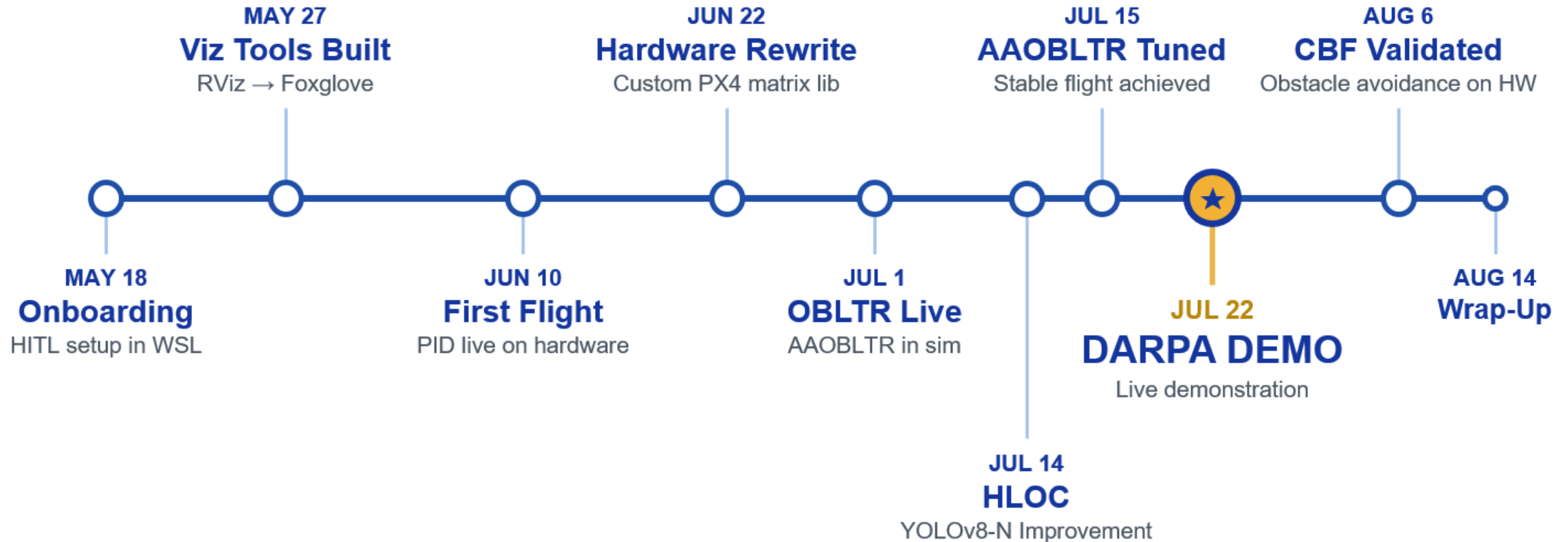


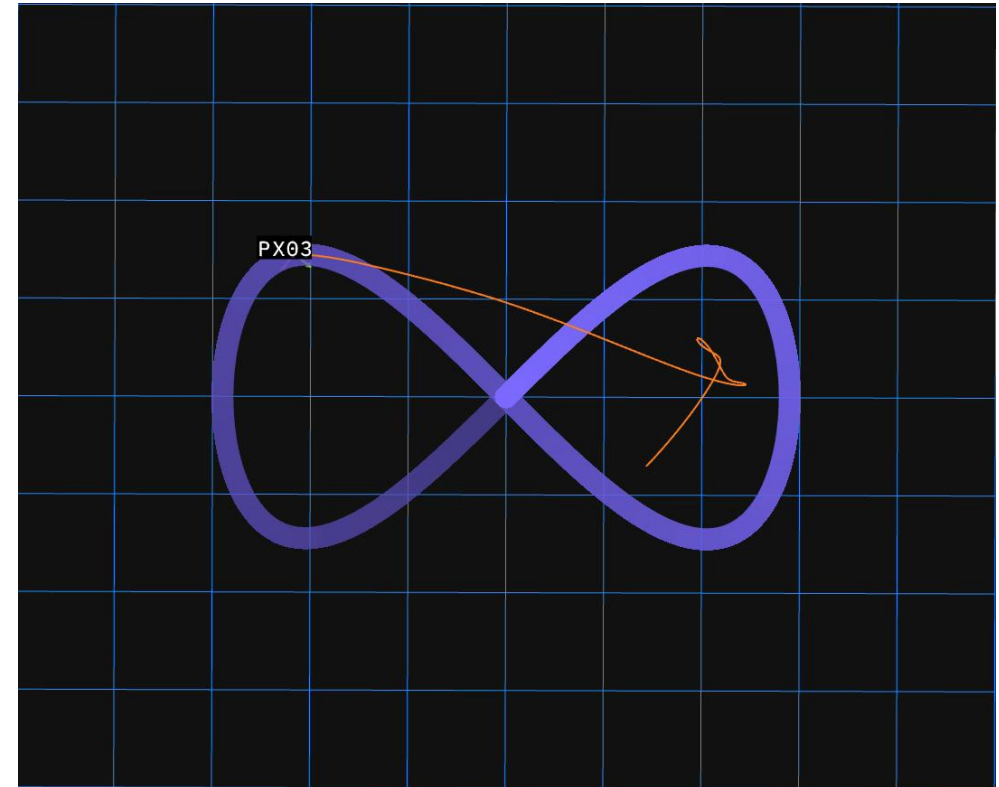
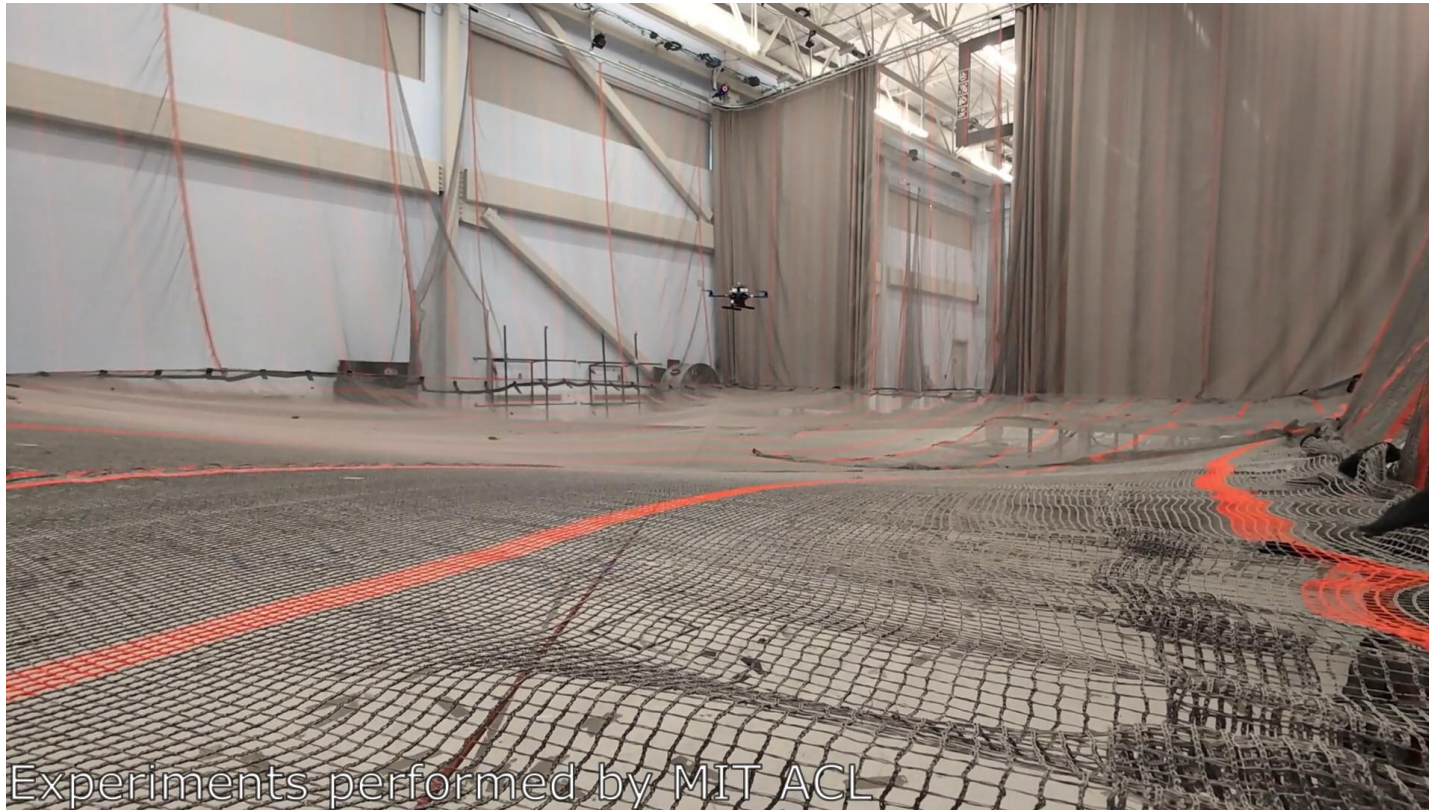
AVI

SCOPE OF MY WORK

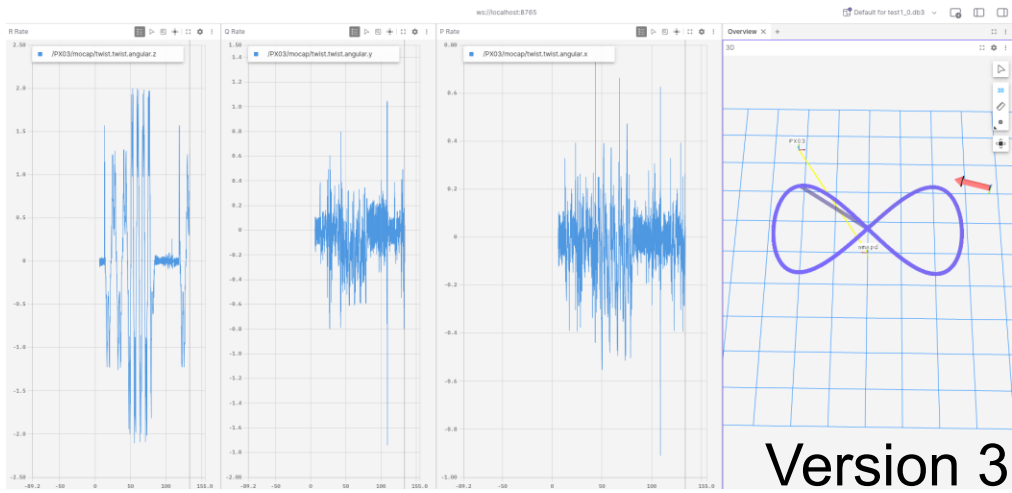
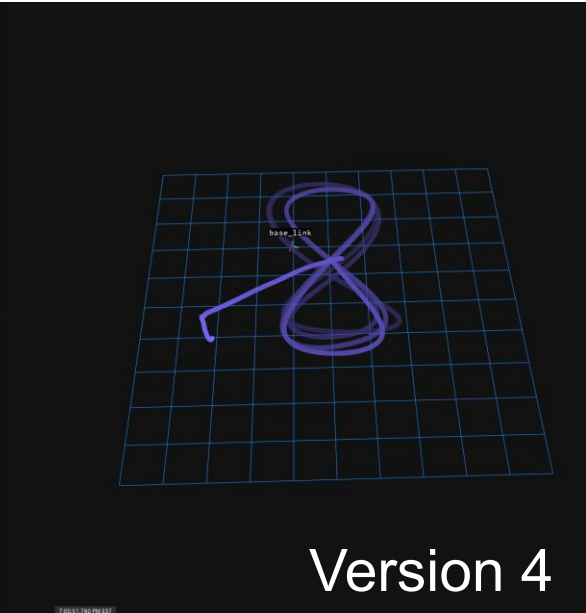
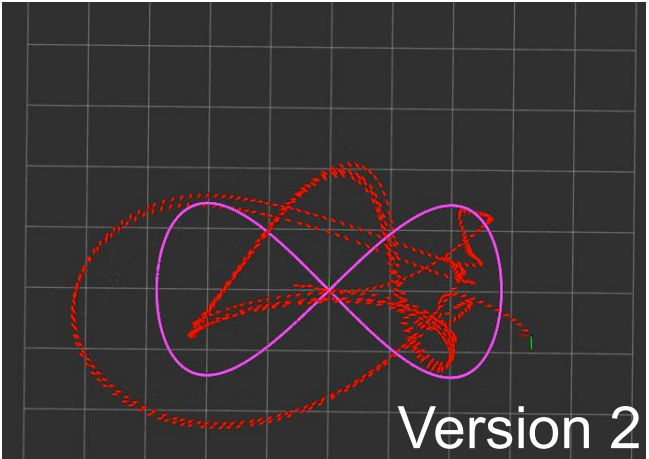
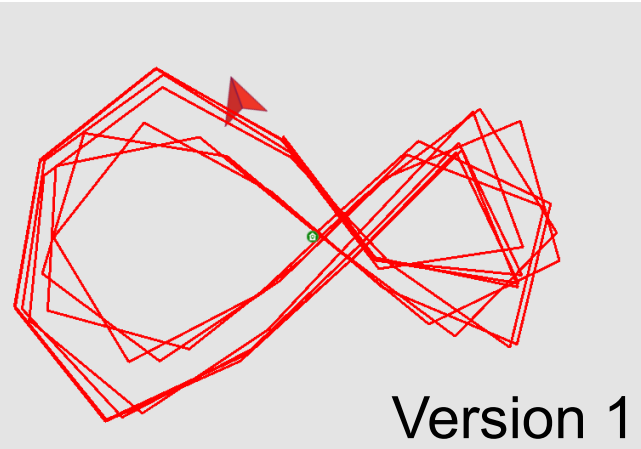
May 18 – Aug 14, 2026 · Aurora Flight Sciences Internship Timeline



DEBUGGING PREVIOUS PX4 IMPLEMENTATION

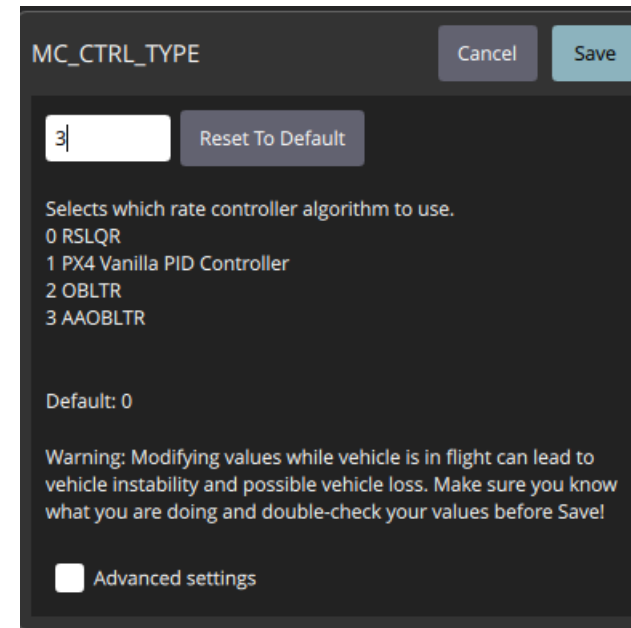
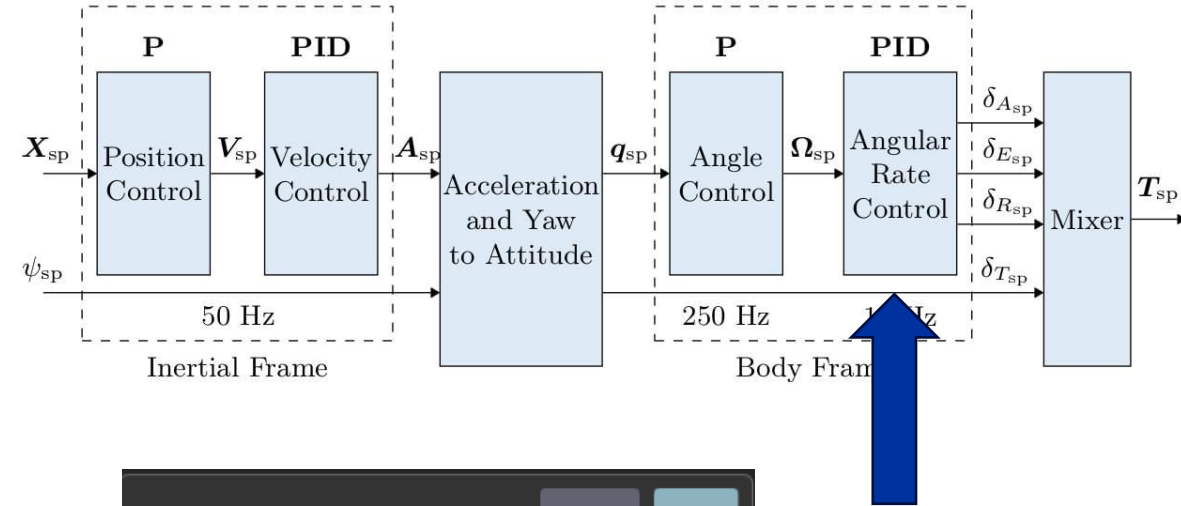


FOXGLOVE VISUALIZATION TOOLS



CUSTOM RATE CONTROLLER LIBRARY IN PX4

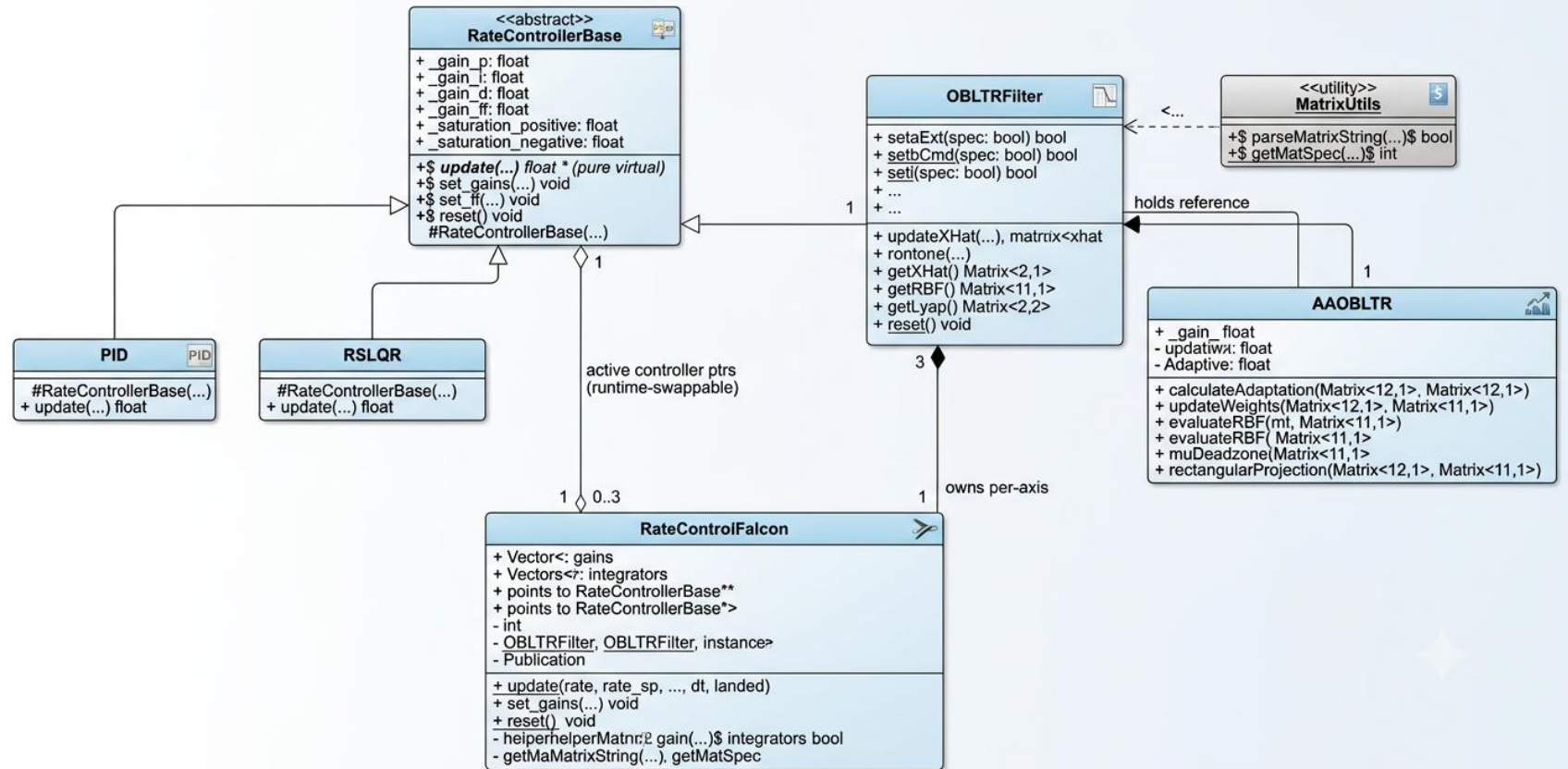
- Implemented in the angular rate control loop
- Forking and built custom PX4 Firmware
- Added new uORB message types
- Extended QGC with custom tuning fields



CUSTOM RATE CONTROLLER LIBRARY IN PX4

- Strategy pattern:
 - Swapable controllers behind one base class
- Four Implementations
 - PID
 - RSLQR
 - OBLTR
 - AAOBLTR
- Runtime-selectable via QGC, no reflash needed

FALCON RATE CONTROL CLASS DIAGRAM



ADAPTIVE AUGMENTED OBSERVER-BASED CONTROL WITH LOOP TRANSFER RECOVERY (AAOBLTR)

- OBLTR:
 - Estimates states the drone can't directly measure
 - Flies as if every state were known
- AAOBLTR:
 - Builds an OBLTR with online self-adjustment
 - Holds performance steady as conditions change

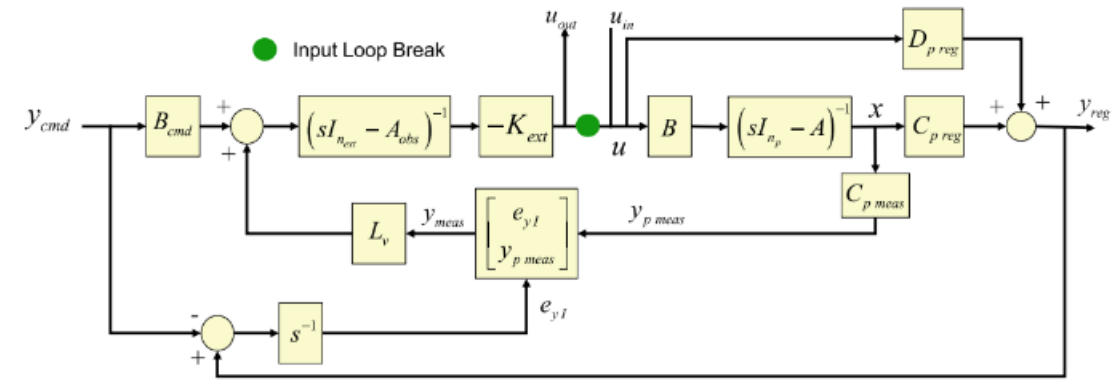
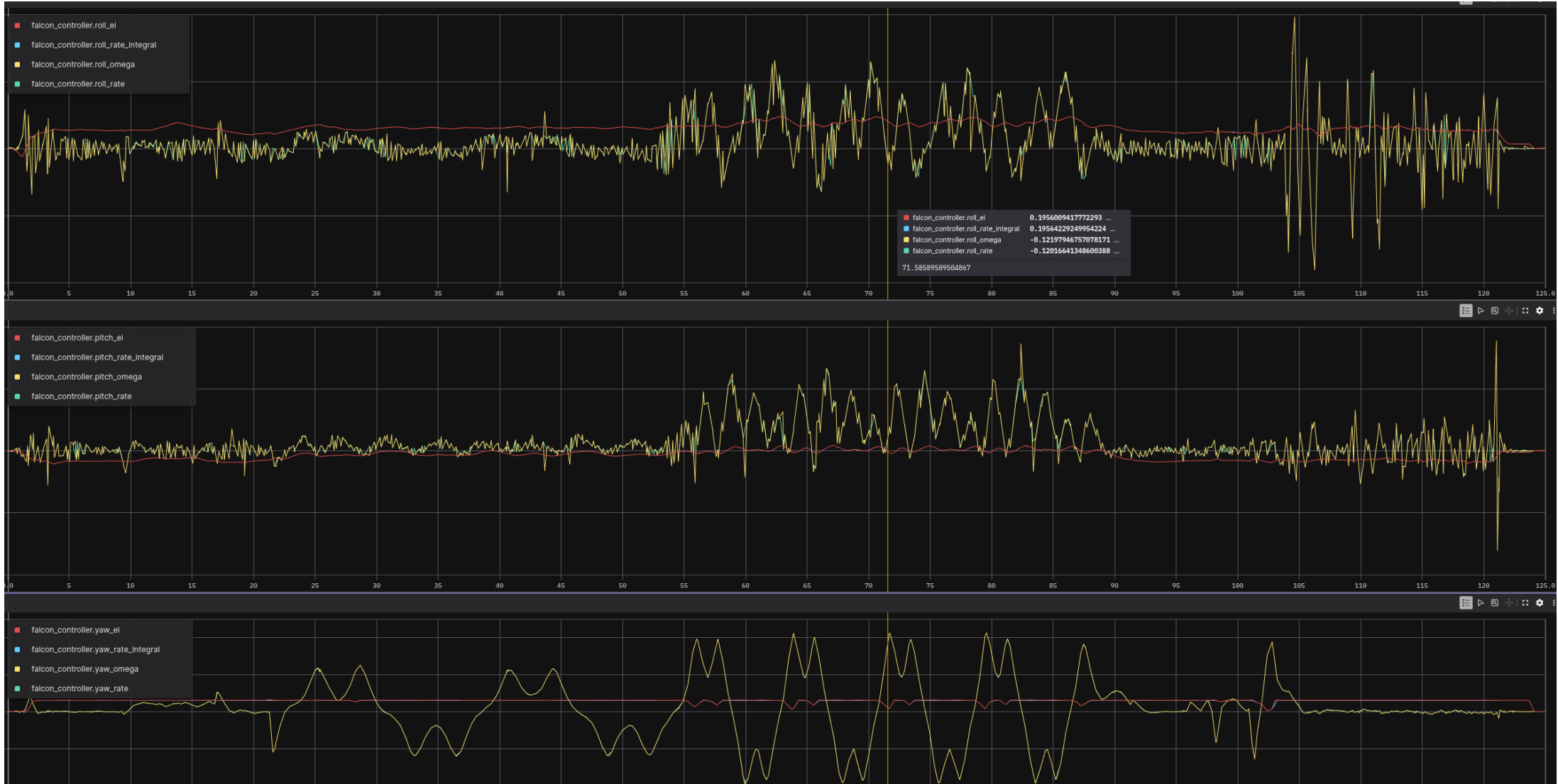


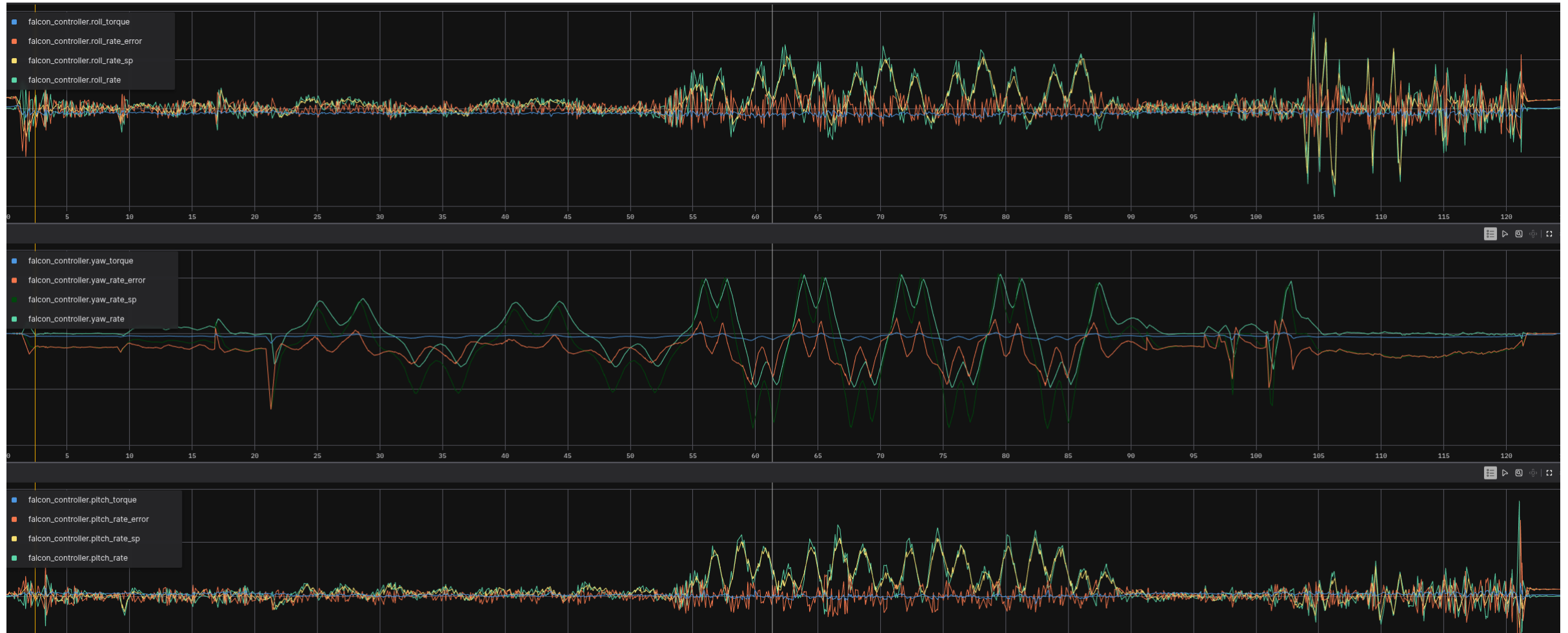
Fig. 6.13 Input loop gain calculation block diagram

TUNING OBLTR / AAOBLTR



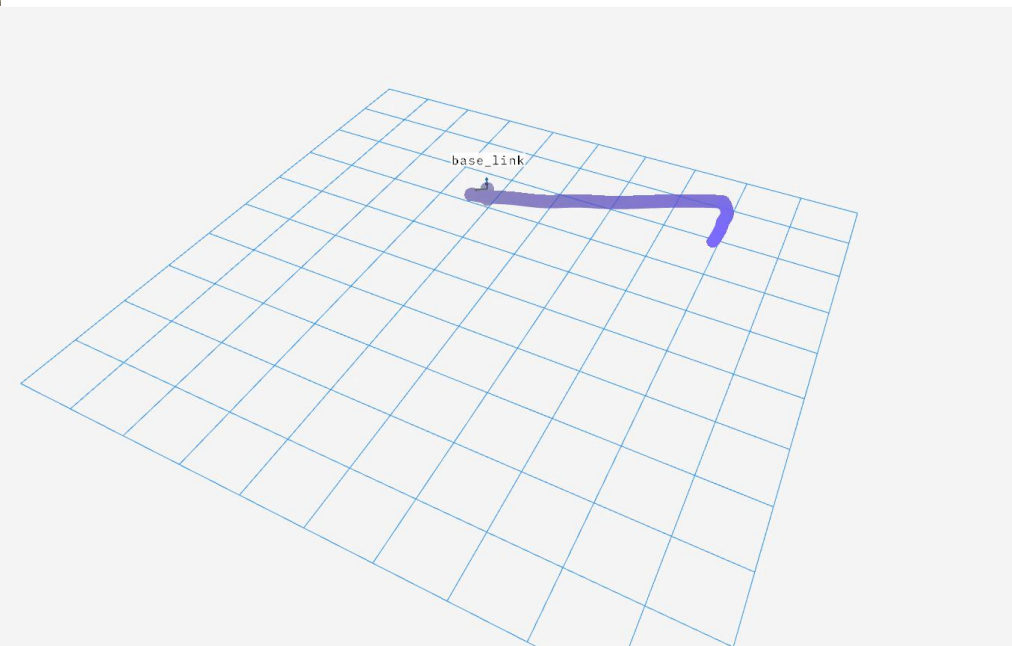
- E_i = Rate Integral
- Ω = Pitch Rate

OBLTR / AAOBLTR TRACKING PERFORMANCE



- Rates are tracking rate setpoints well

AAOBLTR DOING A FIGURE 8



OBSTACLE AVOIDANCE ANALYTICAL CBF SIMULATION

- Closed-form safety filter
- Validated against figure-8 mission with obstacle in path

$$\Delta \mathbf{p} = \mathbf{p} - \mathbf{o}$$

$$h = \|\Delta \mathbf{p}\|^2 - r^2$$

$$\psi_1 = 2 \Delta \mathbf{p} \cdot \mathbf{v} + \alpha_1 h$$

$$\mathbf{c} = 2 \Delta \mathbf{p}$$

$$d = 2 \|\mathbf{v}\|^2 + 2 \alpha_1 (\Delta \mathbf{p} \cdot \mathbf{v}) + \alpha_2 \psi_1$$

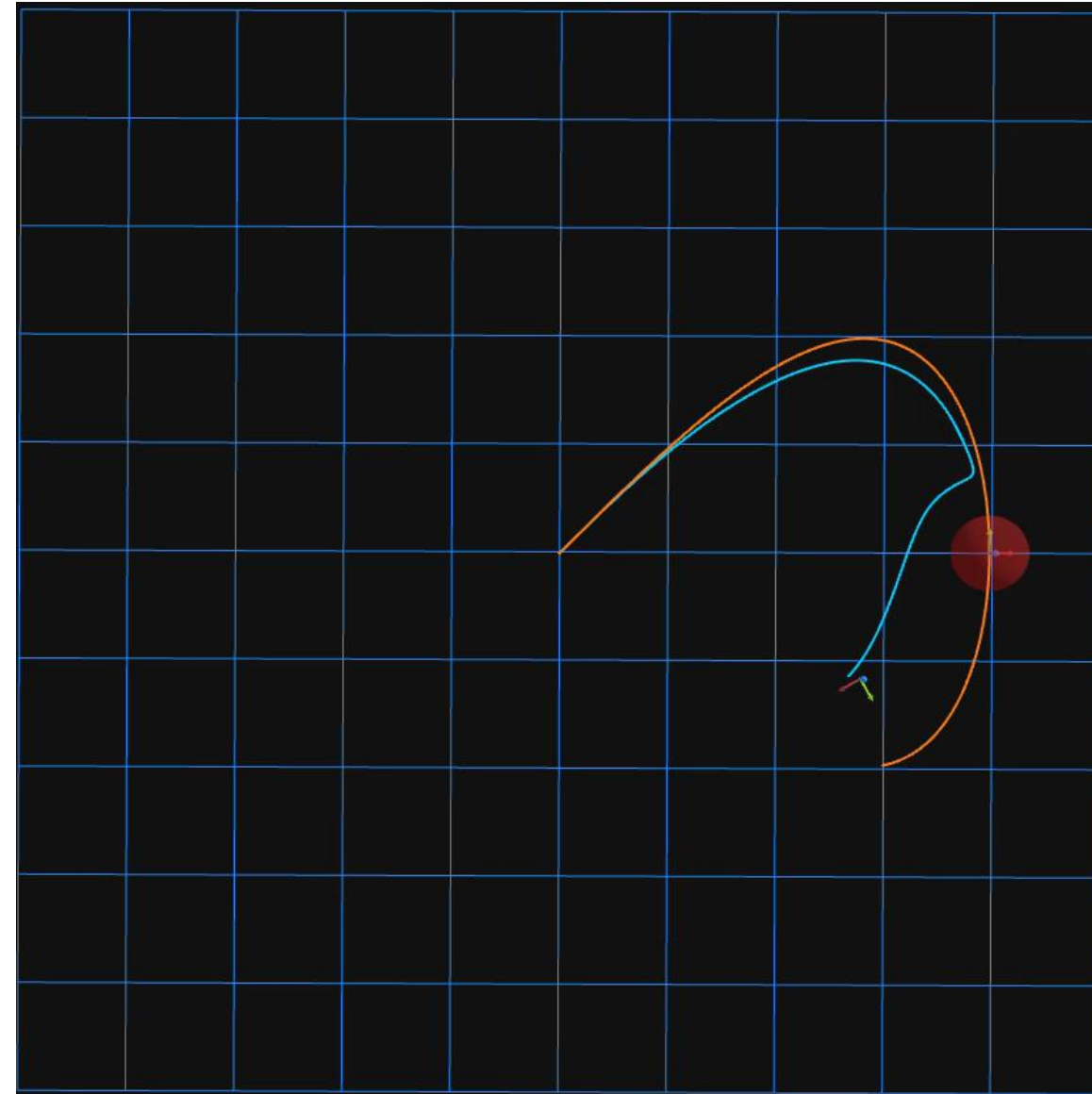
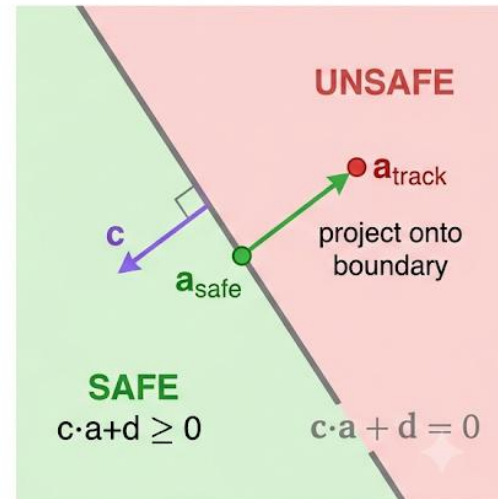
$$\text{margin} = \mathbf{c} \cdot \mathbf{a}_{\text{track}} + d$$

Solution:

$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}}, \text{margin} \geq 0$$

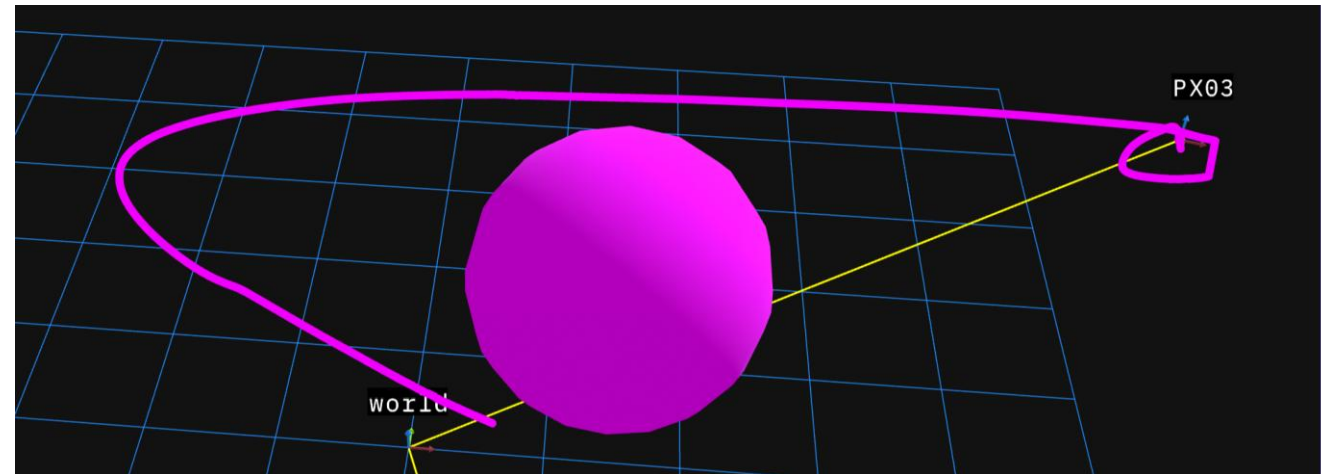
$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}} - \mathbf{c}(\text{margin}/\|\mathbf{c}\|^2), \text{margin} < 0$$

One active obstacle $\Rightarrow$ one linear inequality in $\mathbf{a}$
 $\Rightarrow$ closed form, no QP.



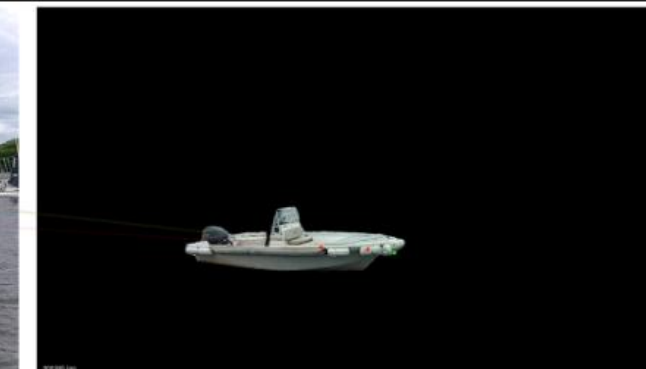
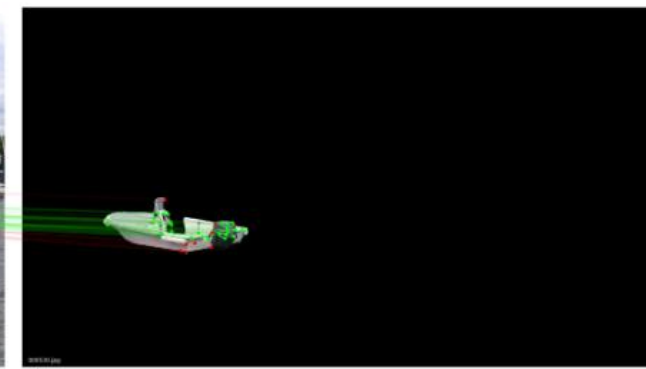
OBSTACLE AVOIDANCE ANALYTICAL CBF HARDWARE

- ROS2 namespace issues blocked goal/CBF message delivery
- Algorithm untested on hardware as a result
- Takeaway:
 - Need a more robust HITL pipeline



HLOC STACK

- Improved HLOC perception stack
 - Inlier count went up significantly
 - Pose estimate accuracy improved
 - Detector: off-the-shelf YOLOv8-n



LESSONS LEARNED

Technical

- Hardware testing is hard
- Steep ramp-up on advanced controllers
- Visualization tools are essential

Professional

- First time with hard deadlines
- Fast-paced, experienced team

People & Culture

- In-office connections matter
- Great work culture boost productivity

